Project: Flappy Bird with PPO

From Data to Solutions
ETH Zurich
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Recap: What you have learned

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Introduction to RL
- Environment, State, Actions

Policy Gradient
- Objective function, REINFORCE

Actor-Critic
- Policy Network, Value Function

GAE & PPO
- Clipped surrogate objective

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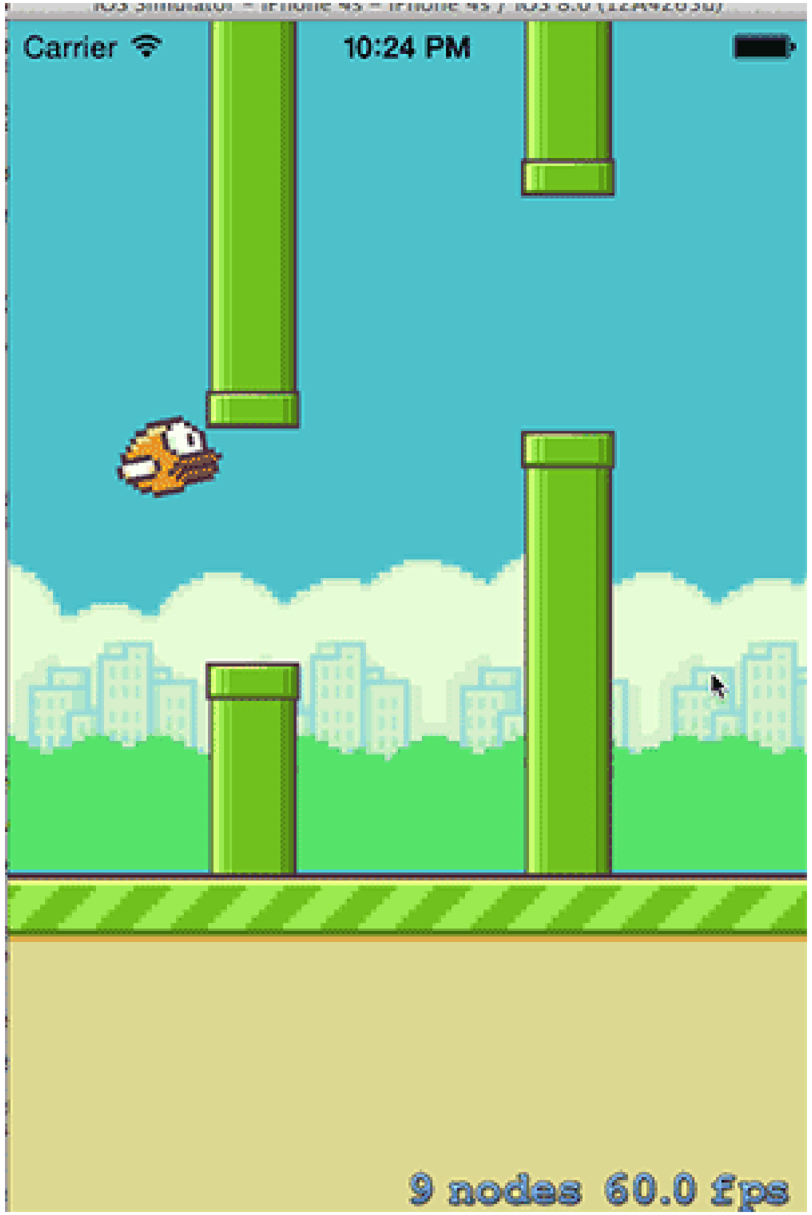
Project Motivation

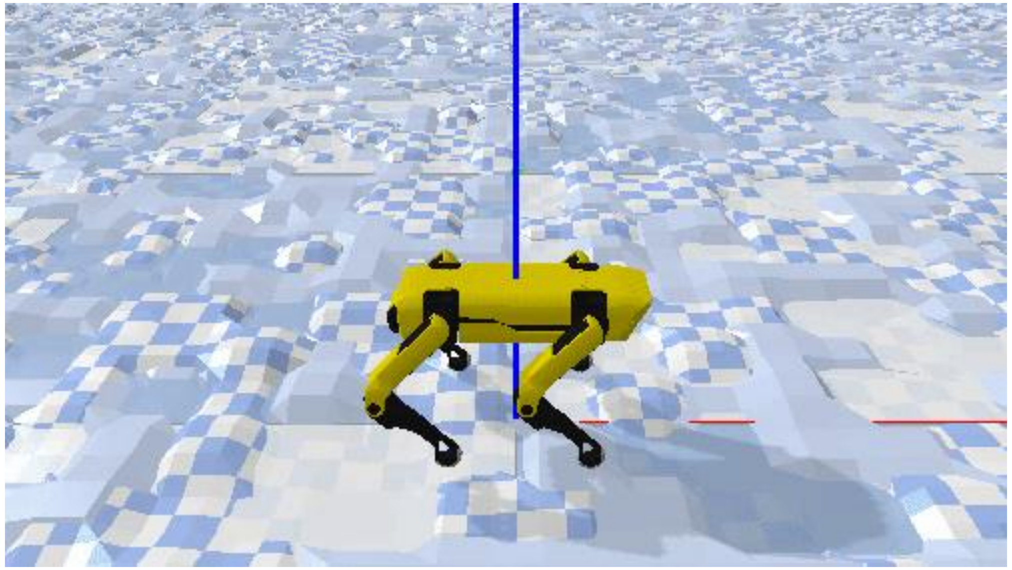
The Game: Flappy Bird

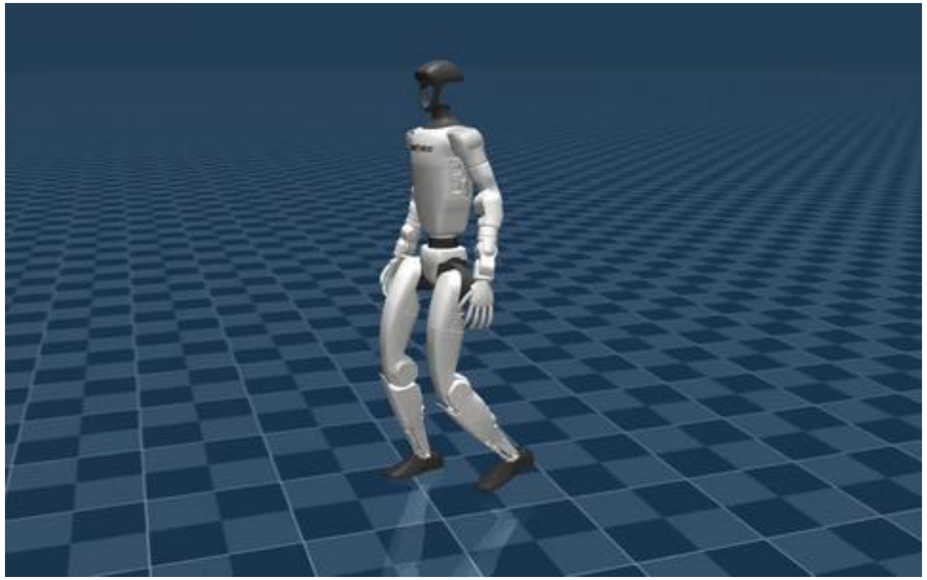
- **Gameplay**: Players must navigate a small yellow bird (Faby) through varying gaps in green.
- **Action**: Players tap the screen to make Faby flap its wings and ascend. If the player stops tapping, gravity pulls the bird down.
- **Objective**: Survival
- **Reward**: Passing through a pair of pipes earns the player one point.
- **Penalty**: A single collision instantly ends the game.



Objective

- Same idea, but this time you *teach* the bird to fly — with PPO — on a clean little discrete-grid version of the game.
- Why?
  - Today, you are training Faby the bird as a single-degree-of-freedom vertical hopper. This serves as a first step to someday train a complex setup, like Spot the robot dog, as a multi-degree-of-freedom quadruped traversing rough and uneven terrain. 😉





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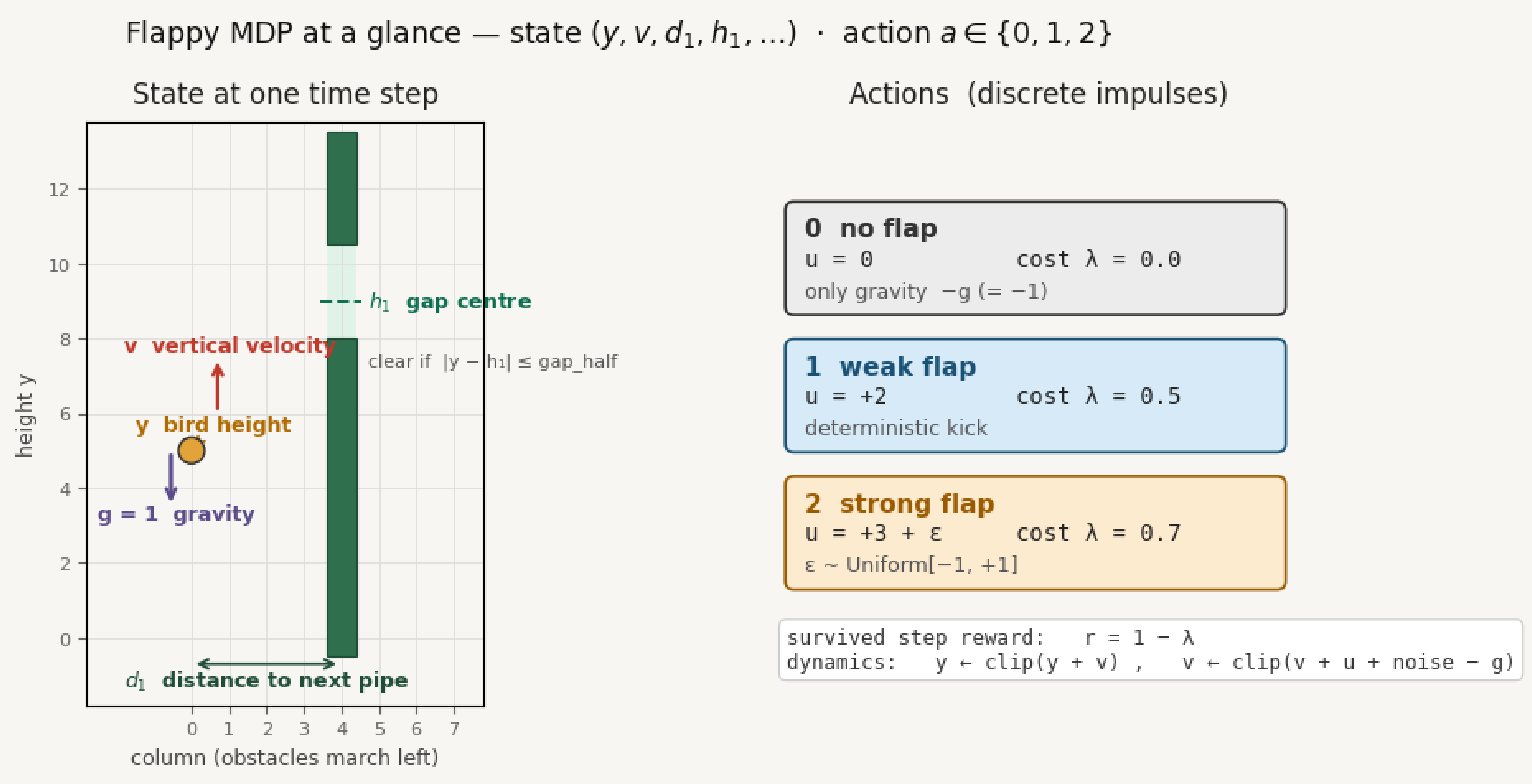
Project Setup & Tasks

Project Setup

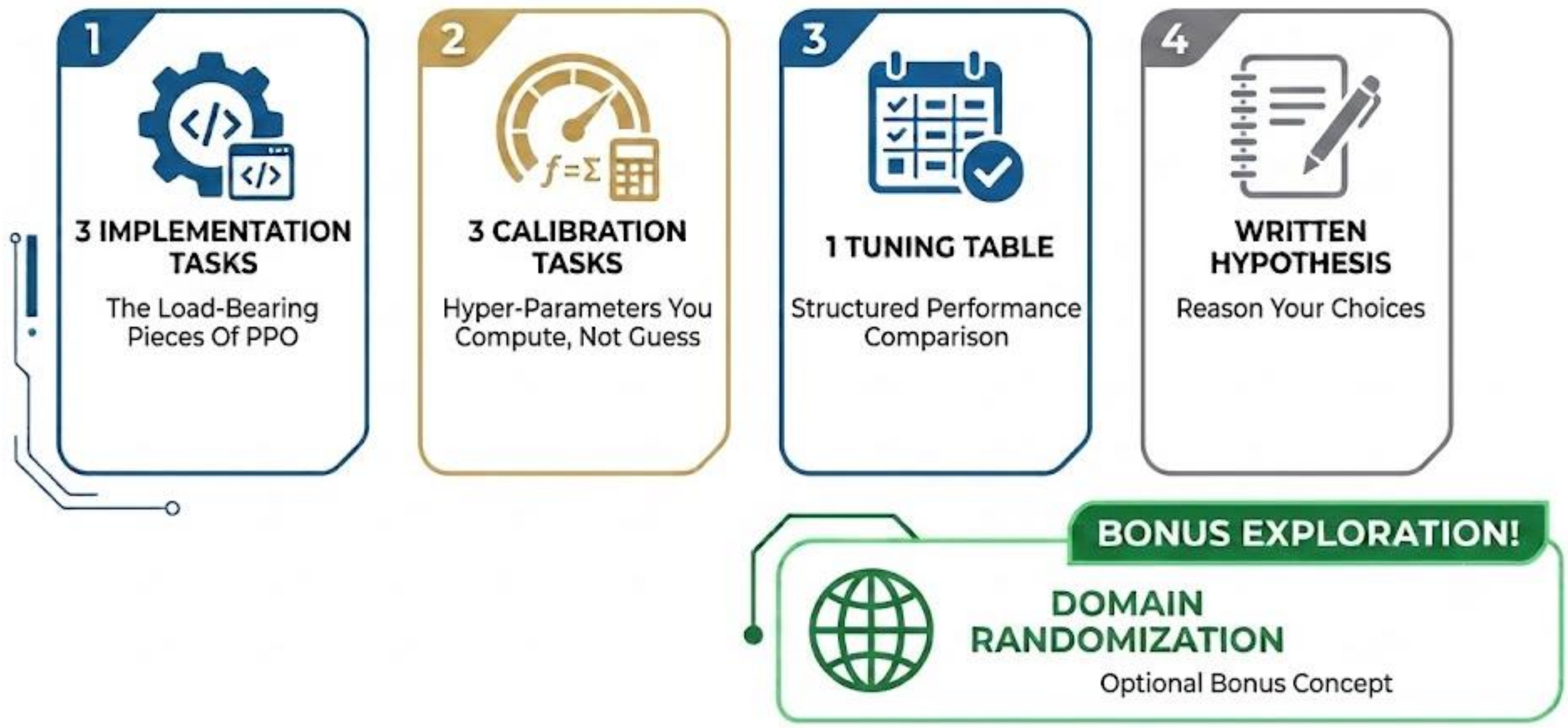


Project Setup

Two Environments

- **Small Instance (Coarse Grid)**
  - State Space: Small enough to fully enumerate.
  - Evaluation: Allows PPO to be compared directly against the true mathematical optimum, rather than a standard baseline.
- **Large Instance (Fine Grid)**
  - State Space: Massive scale (roughly 10^{17} states) sharing the same continuum physics.
  - Evaluation: Requires function approximation (PPO) as the sole tool to learn and evaluate the policy.

Your Tasks



1
3 IMPLEMENTATION TASKS
The Load-Bearing Pieces Of PPO
2
3 CALIBRATION TASKS
Hyper-Parameters You Compute, Not Guess
3
1 TUNING TABLE
Structured Performance Comparison
4
WRITTEN HYPOTHESIS
Reason Your Choices
BONUS EXPLORATION!
DOMAIN RANDOMIZATION
Optional Bonus Concept

Quick Peak: What is Domain Randomization?

Sim-to-Real Gap

Policies trained in a single, static environment often overfit to its specific quirks, causing them to fail when facing variations or real-world noise.

Method

Randomize physical and environmental parameters (e.g., gravity, pipe gaps, wind, friction) at the start of every training episode.

Outcome

The agent is forced to ignore irrelevant noise, learning a highly generalized, robust policy capable of zero-shot transfer to unseen conditions.

Hand In

- Answer all blocks (code and markdowns) indicated with 🎯.
- A training run takes approximately 3-4 minutes. So you have got time for running enough seeds!
- A MP4 file of your trained Faby gets generated at the end so you can visualize your work.
- Submit the completed notebook.

🐦 Enjoy your Project!